AIORI HACKATHON 3 / PROBLEM STATEMENT PS008

Network Management Agent for Reviewed Configuration Changes

A human-approved network change workflow, supported by monitoring and validation evidence from a controlled security lab.

ULTIMATEBLUETEAM | A3-PS008-TC324

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Abstract & Problem Description

THE PROBLEM

Network configuration changes can cause outages or weaken security when changes are applied without consistent review, validation, and rollback.

- Operators need to understand exactly what a proposed change will modify.
- Review should happen before deployment, not after an incident.
- Validation should check configuration syntax, policy, and expected network behavior.
- Monitoring evidence can help verify the impact after a controlled change.

PROPOSE

Agent prepares a structured change and readable diff.

REVIEW + VALIDATE

Human approval and automated checks gate execution.

APPLY + VERIFY

Controlled deployment, post-checks, audit trail and rollback plan.

Scope: a controlled lab—not autonomous changes to production networks.

Background

Why reviewed configuration changes matter in network management

CONFIGURATION STATE

Network devices depend on settings for routing, access control, segmentation, and service availability.

CHANGE RISK

A small unreviewed or invalid change can interrupt connectivity, expose services, or violate security policy.

REVIEW + VALIDATION

A structured diff, explicit approval, pre-checks, and a rollback plan make changes more controlled and auditable.

Where monitoring fits

Suricata and Wazuh can be used as a supporting monitoring layer to inspect security events and help verify unexpected behavior after a change. They do not replace the PS008 agent's core review-and-approval workflow.

Core idea: propose → review → validate → approve → apply → verify → record.

Literature & Standards Review

NETCONF • RFC 6241

A standard protocol for structured network configuration operations. Relevant to controlled device configuration management.

YANG • RFC 7950

A data modeling language for network configuration and state; useful for schema-aware validation.

Suricata + Wazuh

Suricata produces network security events; Wazuh centralizes and displays events for security monitoring.

Design takeaways

Use structured configuration artifacts and reviewable diffs.

Separate authorization from execution and validate before deployment.

Keep monitoring and audit evidence to support troubleshooting and verification.

Proposed Solution & Workflow

01 • INTENT

Engineer describes desired change and target lab scope.

02 • DRAFT

Agent prepares a candidate configuration patch and explanation.

03 • REVIEW

Reviewer checks diff, purpose, scope, and potential risk.

04 • VALIDATE

Run syntax/schema, policy, and lab pre-checks.

05 • APPROVE

Authorized person approves the exact reviewed version.

06 • APPLY + VERIFY

Apply in the lab, run post-checks, collect evidence, rollback if needed.

Fail-safe rule: rejection, failed validation, stale approval, or unexpected post-check result must stop or revert the change.

Optimization Proposed

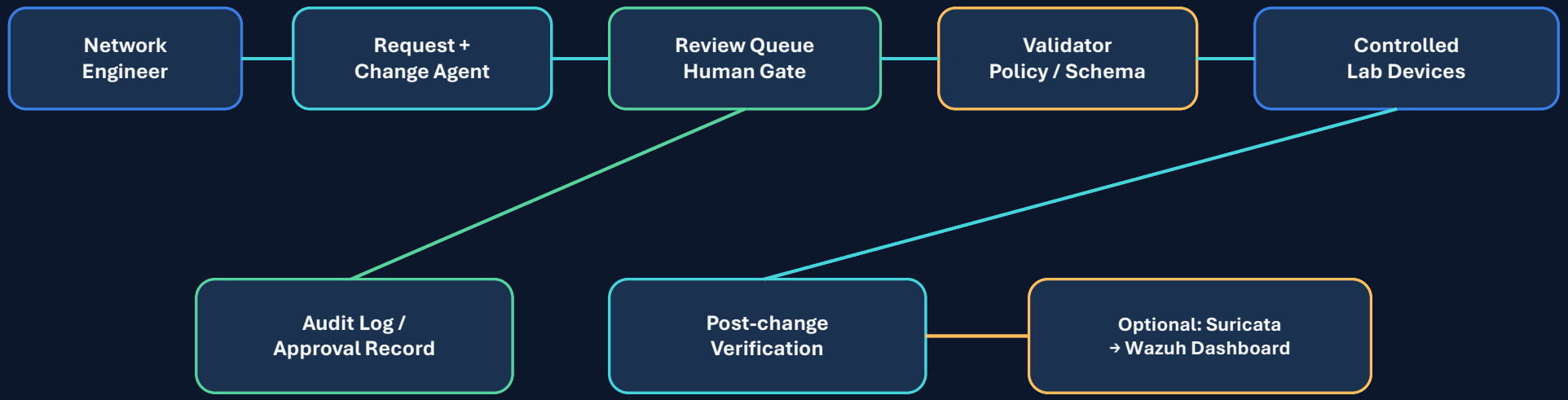
Baseline risk compared with the proposed controlled workflow

AREA	COMMON BASELINE RISK	PROPOSED CONTROL
Visibility	Change intent and actual diff may be unclear	Versioned patch + human-readable explanation
Validation	Manual checks vary between operators	Repeatable syntax, schema, policy and lab checks
Authorization	Approval may be informal or hard to trace	Explicit approval tied to exact patch version
Verification	Impact may not be checked consistently	Pre/post checks and monitoring evidence
Recovery	Rollback steps may be improvised	Documented rollback plan and failure logging

Measure in testing: approval traceability, validation pass/fail, change success, rollback success and time to diagnose.

Solution Architecture & Design

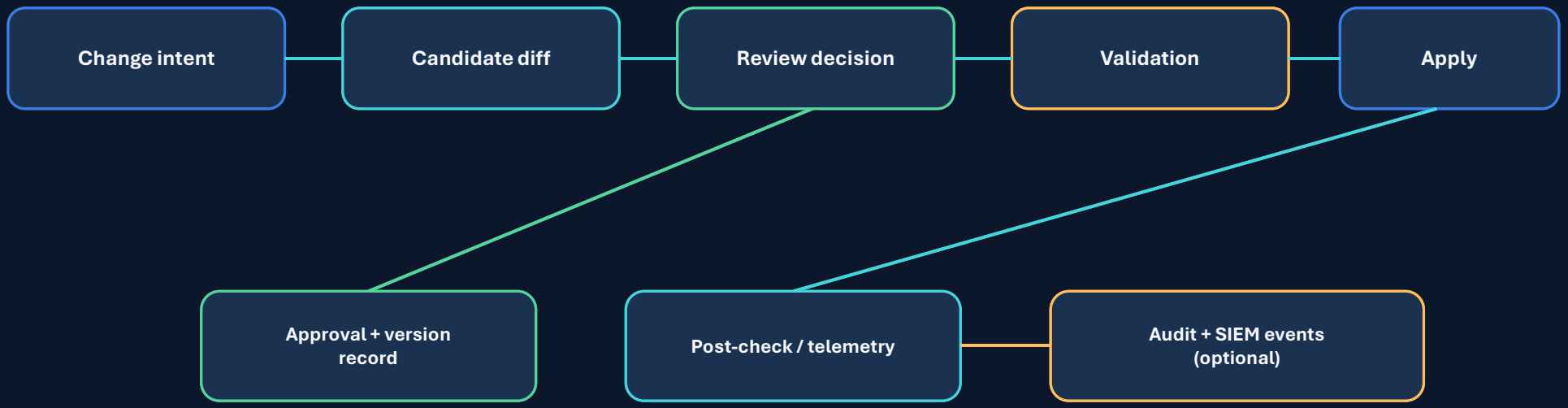
Logical architecture for the proposed PS008 workflow; SIEM components are optional supporting telemetry.



Trust boundary: the agent drafts; a human authorizes; a constrained executor applies only the approved version. Monitoring supports verification but does not replace approval.

Data Flow Diagram (DFD)

Control flow and evidence flow



Gate conditions: no approval → no deployment. Failed validation → revise or reject. Failed post-check → follow rollback procedure and record the outcome.

Deployment Timeline & Methodology

Proposed milestones; replace with actual dates and completed work before submission.

M1	**Scope & threat model**	Choose supported change types, lab topology, roles and safety limits.
M2	**Change request prototype**	Represent intent, patch, diff, reviewer notes and approval status.
M3	**Validation & lab checks**	Implement syntax/policy checks and test safe, rejected and invalid changes.
M4	**Telemetry & verification**	Collect before/after evidence; optionally display Suricata alerts in Wazuh.
M5	**Rollback, audit & demo**	Test failure paths, capture logs, document limitations and prepare demo.

Exit criteria: repeatable demo, negative tests, explicit approval, audit trail and documented rollback behavior.

Bibliography & References

RFC 6241

Enns, R. (Ed.). NETCONF Configuration Protocol. IETF, 2011.
https://www.rfc-editor.org/rfc/rfc6241

RFC 7950

Björklund, M. The YANG 1.1 Data Modeling Language. IETF, 2016.
https://www.rfc-editor.org/rfc/rfc7950

Friend's article

Ansh Srivastav. Integrating Suricata with Wazuh SIEM for Real-Time Threat Detection. Medium.
https://medium.com/@anshsrivastav/integrating-suricata-with-wazuh-siem-for-real-time-threat-detection-98f7e6e9ad53

Wazuh / Suricata docs

Use official documentation to verify log ingestion, event format and lab setup details.
https://documentation.wazuh.com/ and https://docs.suricata.io/

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Thank You

Questions & Discussion

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